



James Stuart (Attempt 1 of 1)

Test Information

QUESTION 1: TRUE/FALSE

5

out of 5 points



The Gibbs free energy of an IDEAL gas is independent of pressure.



Given Answer: False

Correct Answer: False

QUESTION 2: TRUE/FALSE

5

out of 5 points



The Thermal Expansion Coefficient of an IDEAL gas is independent of pressure.



Given Answer: True

Correct Answer: True

QUESTION 3: CALCULATED NUMERIC

5

out of 5 points

The heat capacity of an IDEAL gas at $T=50^\circ\text{C}$ is equal to $C_p^{IG} = 79.314 \text{ kJ} / (\text{kmol K})$. What isthe value of $\left. \frac{\partial S}{\partial \ln T} \right|_V$ for the IDEAL gas. Express your result in **SI** units. Round your numerical answer **into a whole number**.**Hint:** $dS = C_V \frac{dT}{T} + \left. \frac{\partial P}{\partial T} \right|_V dV$

Given Answer: 71

Correct Answer: 71

Answer range +/- 0.1 (70.9 - 71.1)

QUESTION 4: CALCULATED NUMERIC

5

out of 5 points

What is the value of $\left. \frac{\partial S}{\partial \ln P} \right|_T$ for an IDEAL gas. Express your result in **SI** units. Round your numerical answer **into two decimal digits**.



Hint: $dS = C_P \frac{dT}{T} - \beta V dP$

Given Answer: ☒ -8.31

Correct Answer: ☒ -8.31

Answer range +/- 0.01 (-8.32 - -8.30)

QUESTION 5: MULTIPLE CHOICE

5

out of 5 points

An ideal gas is throttled isenthalpically from a pressure $P_1=10$ bar down to a pressure $P_2=2$ bar. The ideal gas temperature change during this process is



Given ☒ Equal to zero

Answer:

Correct ☒ Equal to zero

Answer:

QUESTION 6: TRUE/FALSE

5

out of 5 points

The Heat of Vaporization of a fluid is a monotonically increasing function of the fluid temperature



Given Answer: ☒ False

Correct Answer: ☒ False

QUESTION 7: CALCULATED NUMERIC

5

out of 5 points

The thermal expansion coefficient of a fluid at 70°C is equal to $\beta = 5.71 \times 10^{-4} \frac{1}{K}$. Calculate the numerical value of the derivative $\rho \left. \frac{\partial H}{\partial P} \right|_T$ for this fluid. Round your numerical answer **into two decimal digits**.



Hint: $dH = C_P dT + (1 - \beta T) V dP$

Given Answer: ☒ 0.80

Correct Answer: ☒ 0.8

Answer range +/- 0.01 (0.79 - 0.81)

QUESTION 8: TRUE/FALSE

5

out of 5 points

At equilibrium the chemical potential of the liquid phase is greater than the chemical potential of the vapor phase



Given Answer: ☒ False

Correct Answer: ☒ False

QUESTION 9: CALCULATED NUMERIC

15

out of 15 points

Oxygen is compressed from $P_0=3$ bar and $T_0=30^\circ\text{C}$ to a final pressure $P_F=10$ bar in a single-stage compressor. A thermocouple measures the temperature of the compressor discharge stream to be equal to 165°C . The molar flow rate of oxygen is 8000 kmol/hr and its thermal properties are described by the **IDEAL** gas law. What is the efficiency of the compressor. Express your result as **fraction** and round your numerical answer **into two decimal digits**.



Given Answer: ☒ 0.89

Correct Answer: ☒ 0.89

Answer range +/- 0.01 (0.88 - 0.90)

QUESTION 10: CALCULATED NUMERIC

15

out of 15 points

A steam turbine is used for power generation. Steam is available at pressure $P_1=42$ bar and temperature $T_1=372^\circ\text{C}$. The exhaust pressure is $P_2=1.0$ bar. The turbine efficiency is given by the manufacturer through the following equation $\eta = \frac{|W|}{a + b|W|}$ where $a = 4.23 \times (T_1 - T_2)$ and $b = 1.155 + 0.000538 \times (T_1 - T_2)$. In the above formula T_1, T_2 are in degrees Celsius and W is the power in **kW**. Calculate the power output from the steam turbine at a steam flow rate of $50,000$ kg/hr. Express your result as a positive number in **MW** and round your numerical answer **into two decimal digits**.



Given Answer: ☒ 6.97

Correct Answer: ☒ 6.97

Answer range +/- 0.01 (6.96 - 6.98)

QUESTION 11: CALCULATED NUMERIC

5

out of 15 points

Calculate the **Internal Energy** of ammonia (NH₃) in **kJ/mol** at T=80 °C and P=50 bar. The reference state is Enthalpy $h_{ref}=20.0$ kJ/(mol K) at $T_{ref}=26.9$ °C an $P_{ref}=0.01$ bar. Round your numerical answer **into a single decimal digit**. Assume that in the vapor phase ammonia behaves like an IDEAL gas.



Given Answer: ✖ 26.3

Correct Answer: ✔ 6.6

Answer range +/- 0.1 (6.5 - 6.7)

QUESTION 12: CALCULATED NUMERIC

15

out of 15 points

200 gpm of saturated liquid propane at $T_1=40$ °C are throttled adiabatically down to a pressure of $P_2=1$ bar. Calculate the Lost Work due to the throttling process. Express your result in **kW** and round your numerical answer **into a whole number**. Assume that the vapor phase propane behaves as an **IDEAL** gas. The temperature of the infinite environment is $T_0=300$ K



Given Answer: ✔ 374

Correct Answer: ✔ 374

Answer range +/- 1 (373 - 375)

FEEDBACK AND NOTES FOR ATTEMPT

Feedback to Learner

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

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P 0 WORDS POWERED BY TINY

Grading Notes